RETAIL SALES: DATA AGENTS

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Introduction and Objective

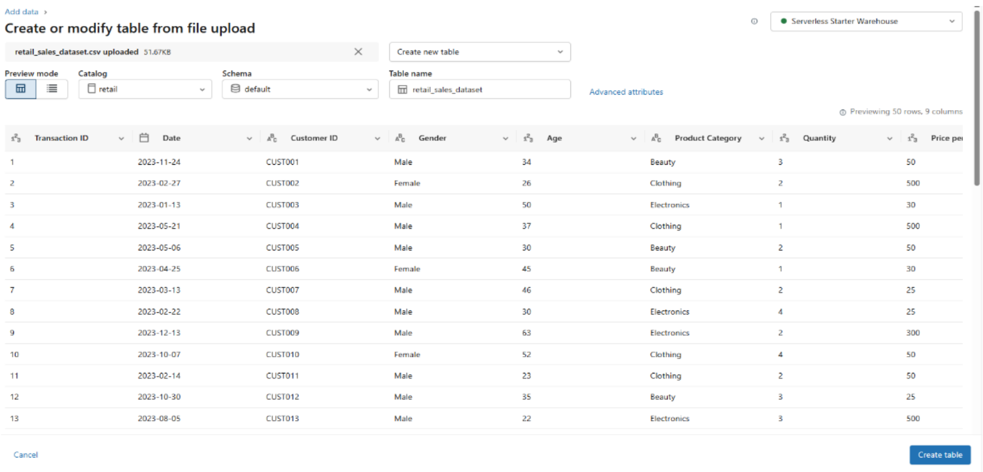
The primary objective of this assignment is to evaluate my ability to apply the knowledge and skills acquired during the BrightLearn Data and AI programme in a practical, real-world scenario. The assignment requires the development of a Retail Sales Data Agent in Databricks that can analyse a retail sales dataset and provide meaningful business insights based on user questions. Through this project, I demonstrate my understanding of data preparation, data exploration, and the implementation of AI-powered data agents. The assignment also assesses my ability to design clear and effective agent instructions that guide the Data Agent to use only the information contained within the provided retail sales dataset. These instructions are important because they define the agent's role, limitations, expected behaviour, and the types of questions it can answer, while preventing it from generating information that is not supported by the data. Overall, the major aim is to apply what Rofhiwa taught in class and produce it on paper, Databricks, and post it on GitHub.

Dataset description

The retail sales dataset contains Transaction ID, Date, Customer ID, Gender, Age, Product Category, Quantity, Price per Unit, and Total Amount. It is a structured dataset that captures when customers made purchases, their gender, age, and the product category of the items they bought. Different customers purchased varying quantities of products at different prices, which together generate total revenue per Customer ID. The dataset can further be explored to determine which gender generated the most revenue and which product category generates the highest revenue.

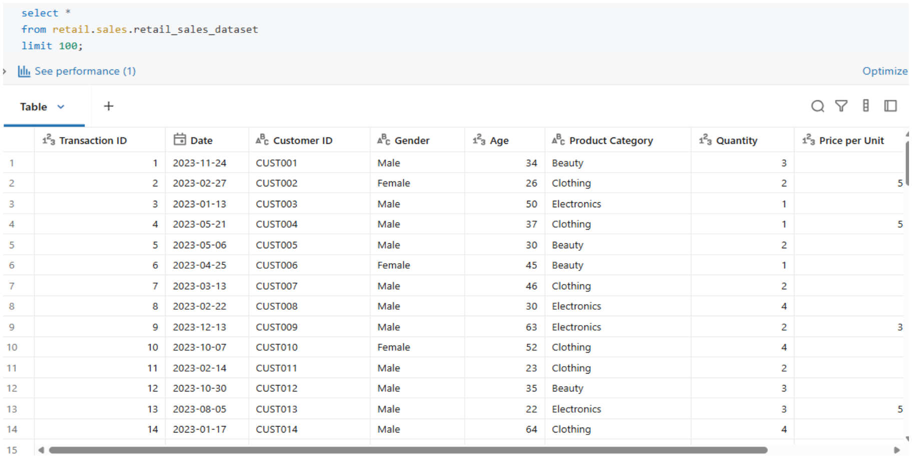
Uploading dataset screenshot, review, table preparations, agent set up and instructions inserted.

Uploaded data onto Databricks by clicking +New on top left corner then, press Add or upload data, then press Create or modify table then dragged the dataset from laptop folder onto Databricks. Then after it immediately showed the below:



Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price pe
1	2023-11-24	CUST001	Male	34	Beauty	3	50
2	2023-02-27	CUST002	Female	26	Clothing	2	500
3	2023-01-13	CUST003	Male	50	Electronics	1	30
4	2023-05-21	CUST004	Male	37	Clothing	1	500
5	2023-05-06	CUST005	Male	30	Beauty	2	50
6	2023-04-25	CUST006	Female	45	Beauty	1	30
7	2023-03-13	CUST007	Male	46	Clothing	2	25
8	2023-02-22	CUST008	Male	30	Electronics	4	25
9	2023-12-13	CUST009	Male	63	Electronics	2	300
10	2023-10-07	CUST010	Female	52	Clothing	4	50
11	2023-02-14	CUST011	Male	23	Clothing	2	50
12	2023-10-30	CUST012	Male	35	Beauty	3	25
13	2023-08-05	CUST013	Male	22	Electronics	3	500

To review the dataset, pressed Query on the mid-left of Databricks, then found this:



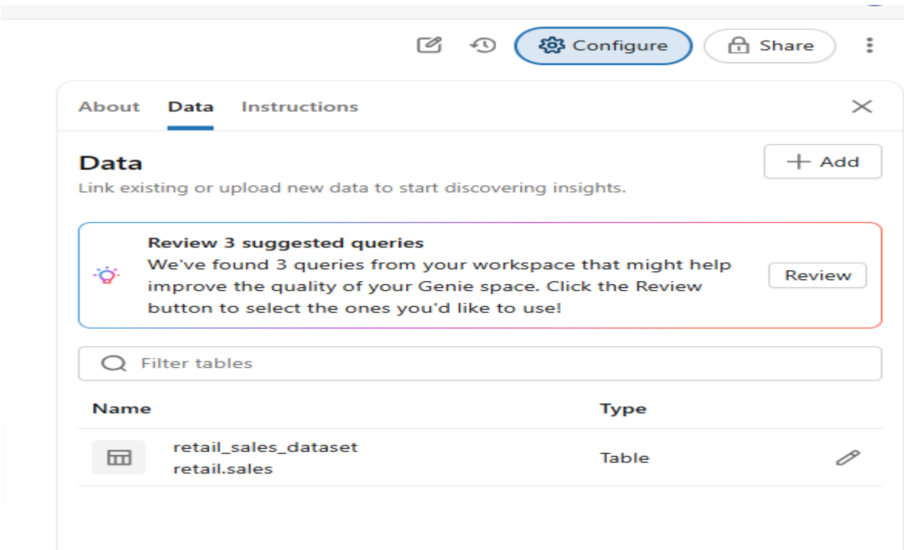
```
select *
from retail.sales.retail_sales_dataset
limit 100;
```

See performance (1)
Optimize

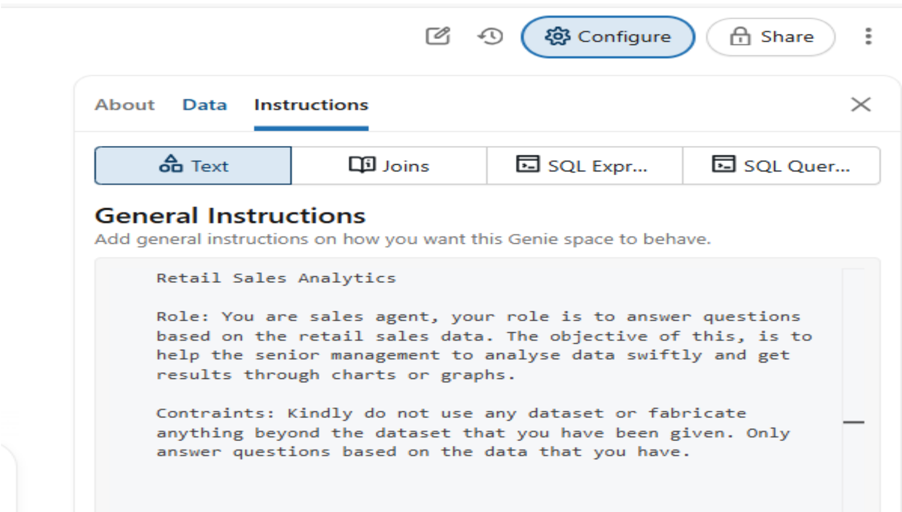
Table

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit
1	1	2023-11-24	CUST001	Male	34	Beauty	3	
2	2	2023-02-27	CUST002	Female	26	Clothing	2	5
3	3	2023-01-13	CUST003	Male	50	Electronics	1	
4	4	2023-05-21	CUST004	Male	37	Clothing	1	5
5	5	2023-05-06	CUST005	Male	30	Beauty	2	
6	6	2023-04-25	CUST006	Female	45	Beauty	1	
7	7	2023-03-13	CUST007	Male	46	Clothing	2	
8	8	2023-02-22	CUST008	Male	30	Electronics	4	
9	9	2023-12-13	CUST009	Male	63	Electronics	2	3
10	10	2023-10-07	CUST010	Female	52	Clothing	4	
11	11	2023-02-14	CUST011	Male	23	Clothing	2	
12	12	2023-10-30	CUST012	Male	35	Beauty	3	
13	13	2023-08-05	CUST013	Male	22	Electronics	3	5
14	14	2023-01-17	CUST014	Male	64	Clothing	4	
15								

To set-up the dataset, retail sales dataset was then added to Genie Spaces. The below shows dataset under Configure, under Data:



Under Configure then Instructions, the below was typed:



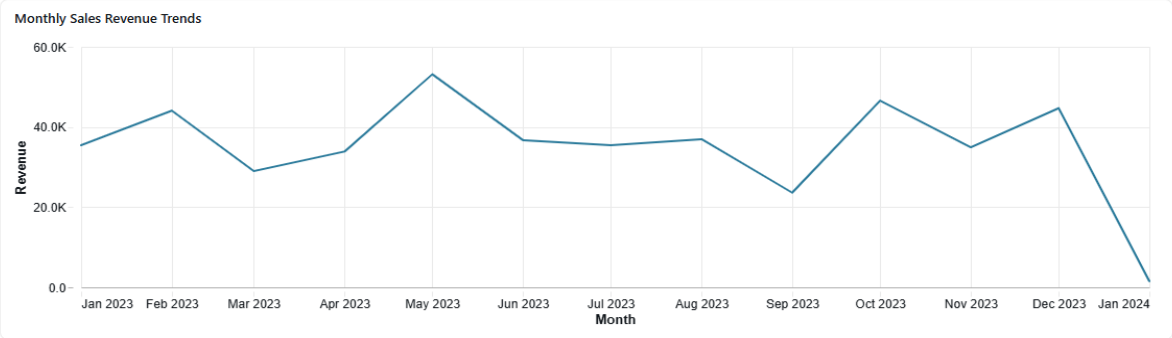
The 11 questions and the answers provided by the agents.

1) Show the monthly sales revenue trends and visualize them through the line graph.

1. Monthly Sales Revenue Trends

Revenue fluctuated throughout 2023, with **May showing the highest sales at $53,150** and **September the lowest at $23,620.** 1

Show code



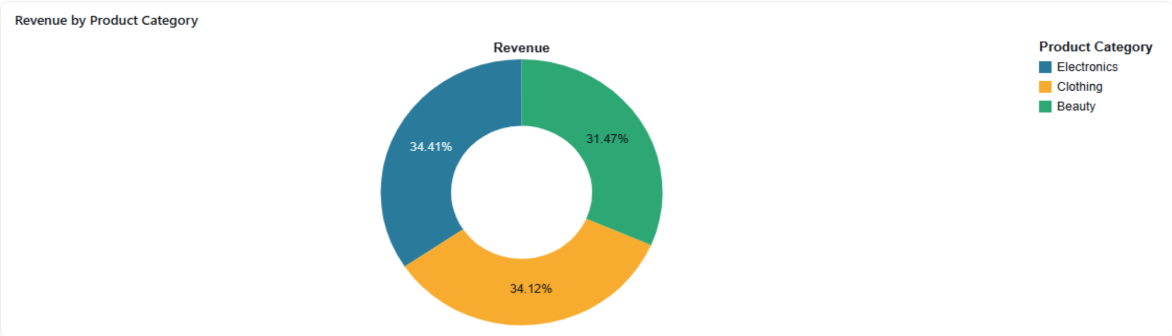
Source: Monthly sales revenue trends

2) Show a pie chart of product categories that generate most revenue.

2. Product Category Revenue Distribution

Electronics generates the most revenue at $156,905 (34.4%), followed by Clothing at $155,580 (34.1%) and Beauty at $143,515 (31.5%). The three categories are remarkably balanced in revenue contribution. 2

Show code



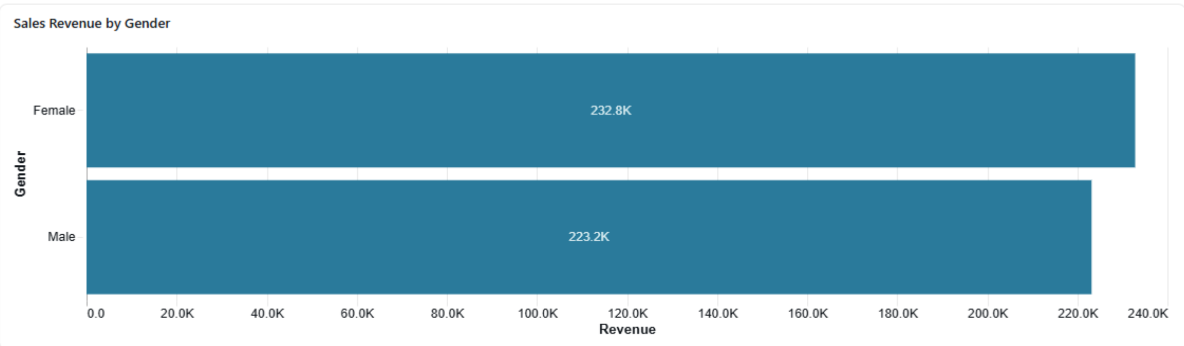
Source: Revenue by product category

3) Compare sales revenue between males and female customers and plot them on the bar chart.

3. Sales Revenue by Gender

Female customers generate $232,840 in revenue (51%), slightly outpacing male customers at $223,160 (49%). The difference is modest at approximately $9,680. 3

Show code



Source: Sales revenue by gender

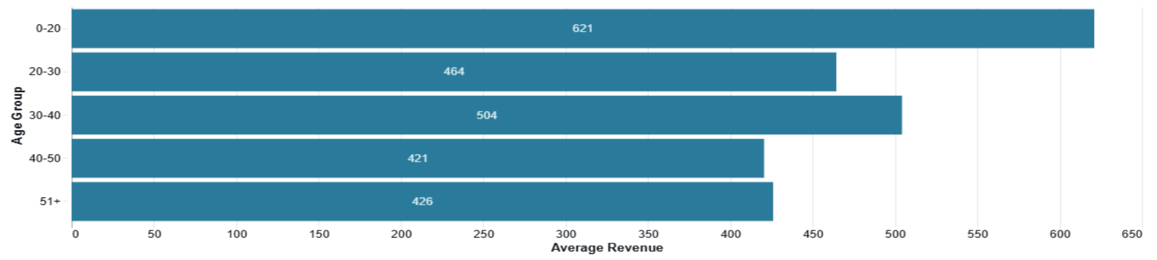
4) Show average sales revenue by age group of 0 and 20, Between 20 and 30, 30 and 40, and between 40 and 50, and lastly 51 and more.

4. Average Sales Revenue by Age Group

The **0-20 age group has the highest average transaction value at $621**, while the 40-50 age group has the lowest at $421. However, younger customers (0-20) represent only 42 transactions compared to 336 transactions from the 51+ group. [4]

Show code

Average Revenue by Age Group



Source: Average revenue by age group

5) Who are the top 5 customers by total spend?

5. Top 5 Customers by Total Spend

Five customers are tied at the top, each spending **$2,000**: CUST072, CUST015, CUST065, CUST074, and CUST089. [5]

Show code

Top 5 Customers by Total Spend

Customer ID	Total Spend
CUST072	$2,000
CUST015	$2,000
CUST065	$2,000
CUST074	$2,000
CUST089	$2,000

Source: Top 5 customers by total spend

6) Which product category sells the highest quantity of units?

6. Product Category with Highest Unit Sales

Clothing sells the most units with 894 units sold, followed by Electronics (849 units) and Beauty (771 units). While Electronics generates the most revenue, Clothing leads in volume. [6]

7) Which days of the week generate the highest sales?

7. Sales by Day of Week

Saturday generates the highest sales at $78,815, followed by Monday ($70,250) and Tuesday ($69,440). Thursday has the lowest sales at $53,835. Weekend shopping (Saturday) significantly outperforms weekdays. [7]

Show code

Sales by Day of Week

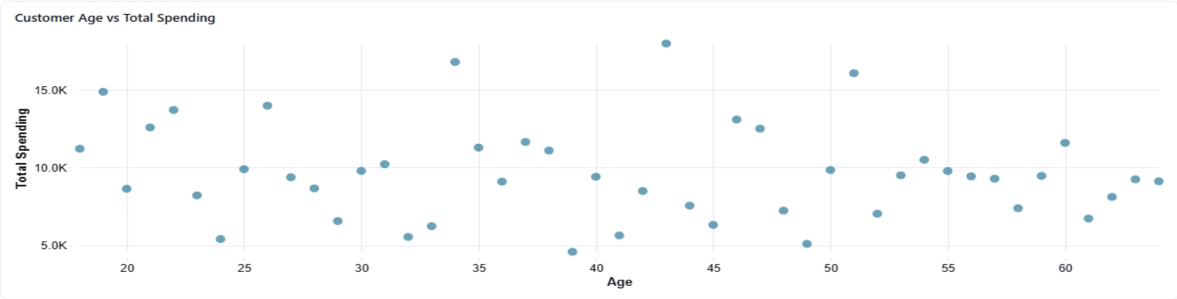
Source: Sales by day of week

8) Show the relationship between customer age and total spending.

8. Relationship Between Customer Age and Total Spending

The scatter plot shows **no strong linear correlation between age and total spending.** Spending is distributed across all age groups, with some concentration of higher spending in the 18-22 and 43-51 age ranges. [8]

Show code



Source: Customer age vs total spending

9) Which product categories are most popular among males and females?

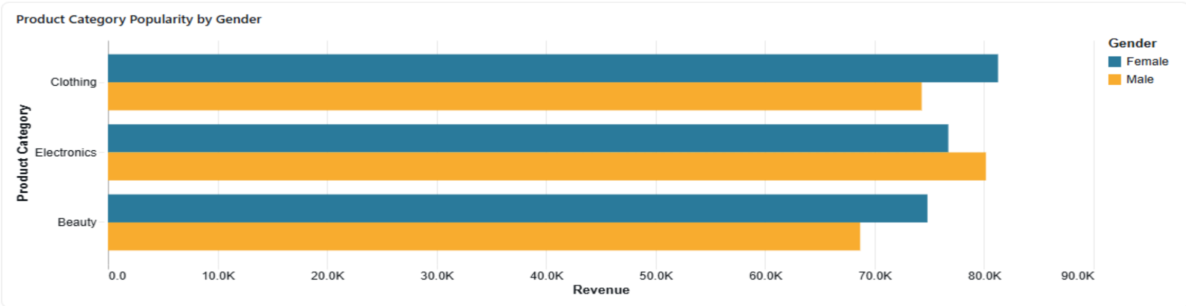
9. Product Category Popularity by Gender

For females: Clothing is most popular ($81,275), followed by Electronics ($76,735) and Beauty ($74,830).

For males: Electronics leads ($80,170), followed by Clothing ($74,305) and Beauty ($68,685).

Key insight: Females prefer Clothing while males prefer Electronics. Beauty is the least popular category for males. [9]

Show code



Source: Product category popularity by gender

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10) Based on historical sales, which product category should management prioritize for future growth and why?

10. Product Category Priority for Future Growth

Recommendation: Prioritize Electronics

Electronics should be the focus for future growth because:

- **Highest total revenue:** $156,905 (34.4% of total)
- **Highest average transaction value:** $458.79 per transaction
- **Strong appeal to male customers:** Males spend $80,170 on Electronics (their top category)
- **Balanced volume:** 849 units sold (only 45 units behind Clothing)

While Clothing sells more units, Electronics generates more revenue per transaction, indicating higher margins and customer willingness to spend. [10]

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11) Based on historical sales, which age group should management prioritize for future growth and why?

11. Age Group Priority for Future Growth

Recommendation: Prioritize the 51+ Age Group

The 51+ age group should be the primary focus because:

- **Highest total revenue:** $143,155 (31.4% of total revenue)
- **Largest customer base:** 336 transactions (33.6% of all transactions)
- **Solid average transaction value:** $426 per transaction
- **Sustainable volume:** This demographic represents the most consistent revenue stream

While the 0-20 age group has the highest average transaction value ($621), they represent only 42 transactions (4.2% of volume). The 51+ group combines strong transaction values with high volume, making them the most valuable segment for sustained growth. [11]

Validation of 3 questions

Validation to question 3,

Date Functions | Retail_Sales_Report | Validations ×

Run all (1000) | 2 minutes ago (29s) | workspace . default | New SQL editor: ON

```sql
SELECT
  `Gender`,
  SUM(`Total Amount`) as total_revenue
FROM
  retail.sales.retail_sales_dataset
WHERE
  `Gender` IS NOT NULL
GROUP BY
  `Gender`
ORDER BY
  total_revenue DESC;
```

Add parameter

Table

	Gender	total_revenue
1	Female	232840
2	Male	223160

This was to analyse which gender generates more revenue and it revealed that females generates more revenue than males. Females buy products more than males and even if males are buying, at times it is for females.

Validation to question 5,

```sql
 SELECT
  `Customer ID`,
  SUM(`Total Amount`) as total_spend
FROM
  retail.sales.retail_sales_dataset
WHERE
  `Customer ID` IS NOT NULL
GROUP BY
  `Customer ID`
ORDER BY
  total_spend DESC
LIMIT 5
```

Add parameter

Table

	Customer ID	total_spend
1	CUST072	2000
2	CUST015	2000
3	CUST065	2000
4	CUST074	2000
5	CUST089	2000

This analysis was to get the top 5 customers who spend the most in the store and the above customers spend the same amount in the store. The analysis was supposed to go further to check which product were they buying. This doesn't tell us much about the spend.

<u>Validation to question 9,</u>

Date Functions | Retail_Sales_Report | Validations

Run all (1000) | Just now (1s) | workspace . default | New SQL editor: ON | Last edit was 1 minut

```sql
LIMIT 5;

SELECT
  `Gender`,
  `Product Category`,
  COUNT(*) as transaction_count,
  SUM(`Total Amount`) as total_revenue
FROM
retail.sales.retail_sales_dataset
WHERE
  `Gender` IS NOT NULL
  AND `Product Category` IS NOT NULL
GROUP BY
  `Gender`,
  `Product Category`
ORDER BY
  `Gender`,
  total_revenue DESC;
```

Add parameter

Table

	Gender	Product Category	transaction_count	total_revenue
1	Female	Clothing	174	81275
2	Female	Electronics	170	76735
3	Female	Beauty	166	74830
4	Male	Electronics	172	80170
5	Male	Clothing	177	74305
6	Male	Beauty	141	68685

The above analysis didn't take out the exact graph on question 9 given that it includes, transaction_count which is not on the graph of the question 9. The primary aim was to see which gender generates more income according to the product category. Overall, the inverted commas were a challenge until I saw it in class.

<u>Key insights</u>

The agent revealed that the shop's product sales generate almost equal revenue on the second question and females generate more revenue than men on beauty and clothing products. Generally in day to day, females do more shopping than males especially on clothes and beauty products because they want to look prettier and more beautiful hence they purchase more than men. It is odd that 51+ individuals generate more revenue than younger ones given that younger ones are more into fashion and beauty products than old ones. The young ones have more tech knowledge than old ones however, the old ones generated more revenue than young ones. It is not an anomaly that Saturday generates more revenue given that on weekdays, people are at work and can do most shopping on weekends especially Saturday than any other days.

<u>Business recommendations</u>

The business should focus primarily focus on 51+ years age group due to the highest total revenue contribution and has the largest customer base. Selling beauty product more also will help the business grow faster as they have the largest revenue contribution compared to the other two categories which were slightly lower. Saturdays hourly operations should be increased because of the high contribution of the revenue per week. Females have the highest average purchases especially on beauty products and clothing compared to male customers. All three products generates good revenue and the business should strengthen their marketing well and position their products in accordance to the target market needs. Conduct survey on different age groups and different genders on how to particularly sell to them, provide their needs as customers and which areas and days are most suitable for them. Open stores in regions where they are preferred and increase the number of hours on days like Saturday where there is most revenue.

Conclusion

Uploading data onto Databricks and reviewing the data was easy because we had done it with Lerato numerous times, and it was also covered in class with Rofhiwa. Watching the videos on data agents from the BrightLearn portal really helped with understanding description, configuration, and how to post questions in Genie Spaces. The most challenging part was validating whether what I was doing was correct. It took time and practice to get through this stage. I would definitely like to practice more and read further on everything we covered in class.